

SNEZHNOGORSK, Russian FederationWMO: **231790**Lat: **68.100N**Lon: **87.767E**Elev: **94**StdP: **100.20**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.2	(c) -42.3	(d) -47.4	(e) 0.0	(f) -44.9	(g) -44.9	(h) 0.0	(i) -42.3	(j) 7.9	(k) -19.4	(l) 7.0	(m) -18.5	(n) 2.0	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	26.2	18.2	23.6	16.8	21.4	15.8	18.9	24.5	17.7	22.4	16.4	20.2	1.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.2	20.8	15.8	11.3	19.8	14.5	10.4	19.1	54.1	24.6	50.4	22.5	46.6	20.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.9	(b) 6.7	(c) 5.9	(e) -46.3	(f) 29.3	(g) 2.5	(h) 2.2	(i) -48.1	(j) 30.9	(k) -49.6	(l) 32.2	(m) -51.0	(n) 33.4	(o) -52.8	(p) 35.0		
			DB	-46.3	29.3	2.5	-48.1	30.9	-49.6	32.2	-51.0	33.4	-52.8	35.0		
			WB	-46.3	19.8	2.5	-48.1	21.1	-49.6	22.2	-51.0	23.2	-52.8	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.8	-27.4	-24.4	-16.0	-8.8	-1.6	10.2	16.2	11.5	5.5	-5.5	-21.0	-22.2
	DBStd	16.20	9.51	8.42	8.88	6.24	4.55	5.29	4.47	3.89	4.32	6.17	7.94	8.70
	HDD10.0	6499	1158	963	807	563	359	63	4	27	146	481	929	999
	HDD18.3	9227	1417	1196	1066	813	617	248	97	213	384	739	1179	1258
	CDD10.0	351	0	0	0	0	0	69	195	75	12	0	0	0
	CDD18.3	37	0	0	0	0	0	4	30	2	0	0	0	0
	CDH23.3	233	0	0	0	0	0	32	200	2	0	0	0	0
	CDH26.7	42	0	0	0	0	0	4	38	0	0	0	0	0
Wind	WSAvg	2.8	2.8	2.9	3.4	3.4	3.1	2.7	2.2	2.4	2.6	3.1	2.5	2.8
Precipitation	PrecAvg	856	70	61	60	59	60	60	64	81	73	108	79	81
	PrecMax	1131	136	109	157	121	100	110	120	142	150	176	142	148
	PrecMin	531	15	16	11	6	21	14	6	28	33	32	16	10
	PrecStd	125	32	25	30	27	21	23	31	26	26	36	33	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.2	-5.1	1.0	5.5	11.6	26.8	29.8	23.1	17.8	8.5	-1.0	-4.1
		MCWB	-7.6	-5.8	-0.2	0.9	5.9	16.3	19.5	17.3	12.9	6.6	-1.2	-4.6
	2%	DB	-10.5	-7.4	-0.4	2.7	8.1	23.4	27.7	21.4	15.2	4.9	-3.4	-6.0
		MCWB	-11.1	-8.0	-1.8	0.2	3.6	15.4	18.9	16.3	11.5	3.5	-3.7	-6.5
	5%	DB	-12.8	-10.4	-1.8	1.3	5.9	20.7	25.7	19.4	13.3	3.0	-5.4	-7.9
		MCWB	-13.3	-11.1	-3.1	-0.5	2.5	14.1	18.2	15.6	10.8	2.0	-5.8	-8.4
	10%	DB	-15.1	-13.0	-3.4	0.1	4.1	18.2	23.5	17.4	11.2	1.6	-9.4	-10.1
		MCWB	-15.6	-13.6	-4.6	-1.3	1.5	13.1	17.1	14.4	8.9	0.5	-9.9	-10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.6	-5.5	0.0	1.9	6.6	18.9	21.1	18.2	13.6	6.7	-1.3	-4.5
		MCDB	-7.2	-5.1	0.9	3.6	10.5	24.2	27.2	21.2	16.4	8.5	-1.2	-4.2
	2%	WB	-11.1	-8.0	-1.6	0.8	4.2	16.3	19.8	17.2	12.4	3.9	-3.6	-6.4
		MCDB	-10.6	-7.5	-0.4	2.3	7.3	21.8	26.0	20.4	14.6	4.5	-3.4	-6.0
	5%	WB	-13.3	-11.1	-3.0	-0.2	2.9	14.7	18.7	16.2	11.0	2.2	-5.9	-8.3
		MCDB	-12.8	-10.5	-1.7	1.2	5.2	19.9	24.5	18.8	13.1	2.9	-5.4	-7.8
	10%	WB	-15.6	-13.6	-4.6	-1.4	1.8	13.4	17.7	14.9	9.4	0.7	-9.8	-10.6
		MCDB	-15.1	-13.0	-3.5	0.1	3.6	17.6	22.7	17.0	11.1	1.4	-9.4	-10.1
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.2	7.6	8.2	6.4	7.7	8.0	6.1	4.6	4.5	5.8	6.7
		MCDBR	6.5	7.6	4.8	8.0	7.8	11.3	10.2	8.6	5.9	4.8	5.6	7.2
		MCWBR	6.3	6.9	4.6	6.6	4.9	6.2	5.1	4.4	3.7	4.2	5.8	7.0
	5% WB	MCDBR	6.5	7.0	4.7	7.3	6.8	10.2	8.9	6.6	5.6	4.7	5.6	7.2
		MCWBR	6.2	6.9	4.5	6.5	5.0	6.5	5.0	3.8	3.9	4.3	5.8	7.2
Clear-Sky Solar Irradiance	taub	0.215	0.256	0.307	0.330	0.360	0.375	0.409	0.395	0.331	0.257	0.199	N/A	
	taud	1.851	1.973	1.998	2.048	2.112	2.349	2.270	2.270	2.389	2.100	1.881	N/A	
	Ebn at Noon	64	505	693	801	820	823	772	735	702	505	64	0	
	Edh at Noon	10	53	95	120	127	105	110	98	67	44	8	0	
All-Sky Solar Radiation	RadAvg	0.03	0.49	2.01	4.13	5.59	5.42	5.05	3.12	1.69	0.58	0.09	0.00	
	RadStd	0.01	0.06	0.22	0.36	0.45	0.48	0.66	0.35	0.18	0.11	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page